

Speaker Notes — ProgressReport_1_merged.pptx (23 slides)

Bilingual, plain-spoken. EN = what you say in English; 中文 = same thing in Chinese.

Handoff: slides 1–7 are the front half (partner); Xiaochuan takes over from slide 8 through the end + appendix.

Slide 1 — Title

EN: Hi everyone. Our project is about a vision-language model — a model that looks at a picture and reads text. We're checking one specific failure: does it pick the image that just *looks* typical, even when that image is actually wrong for the sentence? And the twist we added: does that change depending on which language we ask in — English, Chinese, Arabic, or Hindi.

中文: 大家好。我们做的是个能“看图 + 读文字”的模型。我们想查它一个毛病：它会不会因为某张图“看起来更典型”就选它，哪怕那张图其实跟句子对不上？我们加的新东西是——换不同语言问它（英文、中文、阿拉伯语、印地语），这个毛病会不会跟着变。

Slide 2 — The problem

EN: First, what is this bias. These days we use automatic scorers — things like CLIPScore — to grade AI-generated images instead of asking humans. The problem is they cheat: they give high scores to images that look stereotypical, even when the image doesn't match the prompt. Take the animal example: the prompt only says "an animal near a pond with two bamboo stalks." The correct image has the bamboo but an unusual animal; the wrong image is just a typical dog by a pond. Many scorers still prefer the dog. That's the bias.

中文: 先说这个偏见是什么。现在我们经常用自动打分器（比如 CLIPScore）来给 AI 生成的图打分，省得每次找人。问题是它们会“偷懒”：哪张图看着越像刻板印象，它给的分越高，哪怕那张图根本不符合句子。看动物这个例子：句子只说“一只动物在池塘边，背景有两根竹子”。正确的图有竹子但动物不常见；错误的图就是一只很典型的狗在池塘边。很多打分器还是会选那只狗——这就是偏见。

Slide 3 — Research questions

EN: The original paper only did this in English, and only looked at scoring metrics. Our mentor asked us to add the language angle. So we have two questions. One: does the bias stay the same, or get worse, in other languages — including low-resource ones like Hindi? Two: is the bias stronger for socially sensitive things — like wealth and power — than for neutral stuff like animals and objects?

中文: 原论文只在英文下做，而且只看打分指标。导师让我们加上“语言”这个维度。所以我们有两个问题。第一：换成别的语言——包括印地语这种资源少的语言——偏见是不变还是更严重？第二：在涉及社会敏感的东西上（比如财富、权力），偏见会不会比中性的东西（动物、物体）更重？

Slide 4 — Related work (1/2)

EN: Two papers set up how we read our own numbers. Chen and colleagues showed that when you use an LLM as a judge, the judge brings its own biases — gender, authority, looks. We're using a model as a judge too, so that's exactly what we have to watch. Akyurek and colleagues showed bias measurements are fragile — change the prompt or the metric and the number flips. That's a warning we'll come back to.

中文: 有两篇论文决定了我们怎么看自己的数据。Chen 他们发现：用大模型当"裁判"时，裁判自己就带偏见——性别、权威、长相。我们也是拿模型当裁判，所以这正是要盯紧的。Akyurek 他们发现：偏见的测量很脆弱——换个提示词或换个指标，结论就翻了。这个提醒我们后面会用上。

Slide 5 — Related work (2/2)

EN: Two more. Jin and colleagues built a social-bias benchmark in English and Korean, and found you get different answers depending on how you ask — that's a precedent for testing bias across languages. Hida and colleagues showed bias rankings wobble when you vary the prompt; our different languages are basically prompt variations, and that also flags translation quality as something we have to be careful about. Bottom line: we sit right where judge-bias meets multilingual bias — and we're the first to test this visual bias across languages.

中文: 还有两篇。Jin 他们用英语和韩语做了一个社会偏见测评，发现换个问法答案就不一样——这给"跨语言测偏见"开了先例。Hida 他们发现：换提示词，偏见排名就晃；我们用不同语言其实就是在换提示词，这也提醒我们翻译质量得当心。一句话总结：我们正好卡在"裁判偏见"和"多语言偏见"的交叉点，而且是第一个在多语言下测这种视觉偏见的。

Slide 6 — How we measure

EN: The method is simple. We show the model two images — the correct one and the typical-but-wrong one — and ask, in the target language, "which one matches this description?" We randomly flip which side is left or right, so it can't just always pick the left. Our number is the error rate: how often it picks the wrong, typical image. Half — 0.5 — means it's just guessing, no bias. Above 0.5 means there's bias.

中文: 方法很简单。给模型看两张图——一张正确的，一张"典型但错的"——然后用目标语言问它："哪张符合这句话？"我们会随机左右调换，免得它一直选左边。我们的指标是错误率：它有多大比例选到那张"典型但错"的图。0.5 就是瞎猜、没偏见；高于 0.5 就说明有偏见。

Slide 7 — Illustration: SC vs PA

EN: Here's what the model actually sees. Top row, animals: the correct image is the bamboo with an odd insect; the wrong one is the typical dog by the pond. Bottom row, objects: a scooter versus a typical motorbike. The checkmarks on the right show which scorers fall for the wrong image. Note the prompt never names the object — it only describes the scene. That's the whole game.

中文： 这就是模型真正看到的东西。上面一行是动物：正确的图是竹子加一只不常见的虫子，错的那张是典型的“狗在池塘边”。下面一行是物体：一辆滑板车对一辆很典型的摩托车。右边的对勾表示哪些打分器中招、选了错的图。注意提示词从不说出具体是什么东西，只描述场景——这就是关键所在。

Slide 8 — Setup · (Xiaochuan takes over here)

EN: Okay, now the experiment. The data has three groups: animals, objects, and demography — pictures of people. The demography ones carry an extra tag saying which stereotype they probe: wealth, power, civility, morality, or intellect. Keep that tag in mind — it's the hero of this talk. We ran 300 items per group, times three groups, times four languages — that's 3,600 judgments. Model is Qwen2.5-VL, translations from Google, run on one A40 GPU.

中文： 好，现在进入实验。数据分三类：动物、物体，还有 demography——也就是人物图。人物那类多带一个标签，说明它测的是哪种刻板印象：财富、权力、教养、道德、还是智力。记住这个标签，它是今天的主角。我们每类跑 300 条，三类，四种语言，一共 3600 次判断。模型是 Qwen2.5-VL，翻译用 Google，跑在一块 A40 显卡上。

Slide 9 — Result: the setup

EN: Here's our first look — just average everything by group and language. And honestly? It looks like nothing. Every number hugs 0.5, the gaps between languages are tiny and overlap. At this level, both our questions look flat — no language effect, no domain effect. If we stopped here, the story would be "we found nothing." But we didn't stop here.

中文： 这是我们第一眼看到的——就把每个类别、每种语言简单平均一下。说实话？看着像啥也没有。每个数都贴着 0.5，语言之间差距很小还重叠。在这个层面上，两个问题看着都很平——没有语言效应，也没有类别效应。如果停在这儿，故事就是“我们啥也没发现”。但我们没停在这儿。

Slide 10 — The key move

EN: Here's the trick. Remember that tag on the demography pictures — wealth, power, and so on? The default analysis just lumped all of them into one average. So "demography" became a single number, around .56, and whatever was inside got blended away. Our hunch: the average is hiding the real pattern. So we split it back out by that tag.

中文： 关键的一步来了。还记得人物图上那个标签吗——财富、权力之类的？默认的分析把它们全揉成一个平均值。所以"demography"就变成了一个数，大概 .56，里面真正的东西全被搅没了。我们的直觉是：平均把真实的规律盖住了。所以我们按那个标签把它重新拆开。

Slide 11 — Headline result

EN: And this is the whole report in one slide. Once we split it, that flat .56 blows apart into a huge spread. Wealth: the model picks the stereotype 77% of the time. Power: 67%. But morality and intellect? Right at chance — basically no bias. So the bias isn't everywhere — it's specifically about social *status*. The intuition: the web has a clear look for "rich" or "powerful," but there's no stable look for "moral" or "smart." That's our answer to question two: the bias is way stronger for sensitive attributes than for neutral stuff.

中文: 这一页就是整个报告的核心。一拆开，原来平的 .56 一下子炸成很大的差距。财富：模型有 77% 的时候选刻板印象。权力：67%。但道德和智力呢？就在 0.5 上下，基本没偏见。所以偏见不是到处都有——它专门针对社会“地位”。直觉是：网上对“有钱”“有权”长什么样有清晰的模板，但“有道德”“聪明”没有稳定的长相。这就回答了第二个问题：敏感属性上的偏见远比中性的强。

Slide 12 — Cross-lingual consistency

EN: Now back to the language question. Here's the same breakdown, split by language. And the pattern just... repeats. Wealth and power high, morality and intellect at chance — in English, in Chinese, in Arabic, and in Hindi. The per-language differences are small and within the noise. So the bias isn't coming from any one language's wording — it's baked into the visual side. Language barely moves it; the attribute is what matters.

中文: 现在回到语言这个问题。这是同样的拆分，但按语言分开。结果.....规律就是重复。财富和权力高，道德和智力在 0.5——英文、中文、阿拉伯语、印地语都一样。语言之间的差别很小，都在噪声范围内。所以偏见不是哪一种语言的措辞带来的，它是刻在“图像”这一侧的。语言几乎不影响它，真正起作用的是属性。

Slide 13 — What we learned

EN: Two takeaways. One: the bias is attribute-specific — it lives on social-status attributes like wealth and power. Two: it's basically the same across all four languages. And I want to be clear — "language doesn't change it" is not a boring null result. It actually *strengthens* the original paper: it says this failure is structural, built into the model, not some quirk of English.

中文: 两个结论。第一：偏见是“属性特定”的——它集中在财富、权力这种社会地位属性上。第二：四种语言下基本一样。我想强调一点——“语言不影响它”不是个无聊的“没结果”。它反而加固了原论文：说明这个毛病是结构性的，是模型本身的，不是英文特有的偶然。

Slide 14 — Limitations

EN: Being honest about the weak spots. The effects are real but our error bars are wide — wealth only has about 31 items per language, so we're underpowered for fine language comparisons. It's one model, one seed, one translation tool. We haven't yet checked the Arabic and Hindi translation quality, which is a confound for any language claim. And we haven't audited position bias or how often parsing failed per language. We're naming these before anyone asks.

中文: 老实说说短板。效应是真的，但误差棒很宽——财富每种语言只有大概 31 条，所以做精细的语言对比时样本不够。这是单一模型、单一随机种子、单一翻译工具。阿拉伯语和印地语的翻译质量我们还没校验，这对任何“语言”结论都是个干扰因素。还有位置偏好、各语言的解析失败率也还没查。这些我们主动先说，免得被问。

Slide 15 — Outlook

EN: What's next — and none of it needs a GPU first. Step one: run a proper statistical model to confirm "attribute matters way more than language" is real, not luck. Step two: collect more wealth and power items to tighten those error bars. Step three: add more low-resource languages to directly test the "low-resource is worse" idea. Step four: validate the translations by translating back and checking drift. And here's the nice part: either way we win — if it stays flat, that's language-invariance; if a low-resource language spikes, we've recovered the authors' original hypothesis.

中文: 下一步——而且一开始都不用 GPU。第一：跑一个正经的统计模型，确认“属性比语言重要得多”是真的、不是运气。第二：多收集财富和权力的样本，把误差棒收窄。第三：再加几种低资源语言，直接测“低资源更糟”这个想法。第四：把翻译再翻回去、看偏差多大，来验证翻译质量。最妙的是：怎么样我们都赢——如果还是平的，那就是“语言无关”；如果某个低资源语言冒尖了，我们就证实了原作者的假设。

Slide 16 — Wrap-up

EN: To sum up: we built and ran the first multilingual version of this benchmark — 3,600 judgments across four languages. We found the bias is attribute-specific, concentrated on wealth and power, and consistent across languages — and we only saw it because we caught the standard averaging hiding it. And we've laid out a concrete, doable plan for experiment two. That's it from us.

中文: 总结一下：我们搭起并跑通了这个测评的第一个多语言版本——四种语言、3600 次判断。我们发现偏见是“属性特定”的，集中在财富和权力上，而且跨语言一致——而我们能看到它，只是因为我们抓到了“默认平均把它盖住了”。我们还给第二个实验定了一个具体、能落地的计划。我们就讲到这儿。

Slide 17 — Thank you

EN: Thanks for listening — happy to take questions. The code's in the repo if you want the details.

中文: 谢谢大家——欢迎提问。想看细节的话代码都在仓库里。

Slide 18 — Appendix divider

EN: (Don't present. Just say if asked:) I've got some backup slides for the details — let me pull the relevant one up.

中文: (不主动讲，被问到再说:) 细节我准备了一些备用页，我调出对应的那张。

Slide 19 — Backup B1: bias by socio-attribute (pooled)

EN: This is the full attribute table pooled across languages, with the gender split at the bottom — males .573, females .558, though we haven't tested that difference yet. Happy to walk through any row.

中文: 这是把四种语言合起来的完整属性表，底下是性别拆分——男性 .573，女性 .558，不过这个差异我们还没做检验。哪一行想细看都可以。

Slide 20 — Backup: translation sanity check

EN: This is for the obvious question — "how do you know the translations are any good?" Here's one item in all four languages. Chinese I verified myself since I read it. For Arabic and Hindi, the plan is back-translation checking, which is in our next steps. We're not claiming they're perfect — we're claiming we know it's a risk and we're handling it.

中文: 这一页是为了回答那个一定会被问的问题——"你怎么知道翻译靠谱?" 这是同一条样本的四种语言版本。中文我自己看得懂，亲自核过了。阿拉伯语和印地语，计划用"回译核对"，已经列进下一步。我们不是说翻译完美，而是说我们知道这是风险，并且在处理。

Slide 21 — Backup: full socio_attr × language table

EN: This is the number behind slide 12 — every attribute, every language. The thing to notice: wealth and power, shaded, sit high in *all four* columns; morality and intellect sit near .5 in all four. The order of attributes doesn't change across languages — which is exactly why we say it's the attribute, not the language.

中文: 这是第 12 页背后的数字——每个属性、每种语言。要看的点是：财富和权力（带底色）在四列里都很高；道德和智力在四列里都贴着 0.5。属性的排序不随语言变——这正是我们说"是属性，不是语言"的依据。

Slide 22 — Backup: bias by sub-population

EN: Let me be precise here, because this cut is easy to overclaim. In these demography items the CORRECT image is the one showing a marked identity — a pride flag, a kippah — and the trap image is the plain, unmarked "default" person. On the sexual-orientation items the model picks that unmarked default about 75% of the time, even though it's the wrong answer; religion and nationality are at chance. And this same effect is what inflates the wealth number — wealth-by-sexual-orientation is .97. The honest caveat: those pride markers are visually loud, so we can't yet separate "the model avoids LGBTQ-coded images" from "the model avoids the busier image." Exp2 adds sub-population and the visual knob to the model to pull those apart. And to be clear — we're measuring the model's bias, not endorsing any stereotype, and we only show numbers.

中文: 这里我要讲准一点，因为这个切片很容易被过度解读。在这些 demography 样本里，"语义正确"的那张图恰好是带可见少数群体标记的（彩虹旗、犹太基帕），而"典型陷阱"图是那个没有任何标记的"默认人"。在性取向这一类上，模型大约 75% 的时候选了那个无标记默认人，哪怕它是错的答案；宗教和国籍则在随机线上。而且正是这个效应把财富那个数顶高了——财富 × 性取向那格是 .97。要诚实说的 caveat：那些彩虹标记在视觉上很显眼，所以我们目前分不开"模型在回避 LGBTQ 相关的图"和"模型只是在回避更花哨的图"。Exp2 会把人群类型和视觉旋钮一起放进模型来拆开这两者。还有，我们是在测量模型的偏见，不是认同任何刻板印象，而且只展示数字。

Slide 23 — Backup: object × Hindi

EN: One interesting exception worth flagging. On objects, English, Chinese, and Arabic all sit at .46-.47 — no bias. But Hindi alone jumps to .55. It's the one cell where the low-resource language behaves differently — which is a little hint toward "low-resource is worse." But I won't over-sell it: the sample's small and the confidence interval still touches 0.5. It's a lead to chase in experiment two, not a result.

中文: 有个有意思的例外值得提一下。在物体类别上，英文、中文、阿拉伯语都在 .46-.47——没偏见。但只有印地语跳到 .55。这是唯一一个低资源语言表现不一样的格子——有点像"低资源更糟"的苗头。但我不会夸大：样本小，置信区间还碰着 0.5。它是第二个实验要追的线索，不是结论。

Slide 24 — Backup: confounds & ethics

EN: Two honest points. First, a confound: the wrong image is made by turning one visual knob — count, color, layout, and so on — so a wrong pick might track that low-level change, not "typicality" itself. Our next-step model will control for it. Second, ethics: the demography data encodes race, religion, and orientation stereotypes — sensitive stuff. So in public slides we only show animal and object images; demography stays as labels and numbers. We study the stereotype to measure it, never to endorse it.

中文: 两个诚实的点。第一，一个干扰因素：那张错的图是靠改一个视觉旋钮做出来的——数量、颜色、布局之类——所以模型选错可能是跟着这个低层变化走，不一定是"典型性"本身。我们下一步的模型会把它控制住。第二，伦理：人物数据里带着种族、宗教、性取向的刻板印象，很敏感。所以公开幻灯里我们只展示动物和物体图，人物只用标签和数字呈现。我们研究刻板印象是为了测量它，绝不是认同它。